

Gaiye “Gail” Zhou, MBA, PhD

LinkedIn: www.linkedin.com/in/gailzhou Email: gail.j.zhou@gmail.com

Principal Software Engineer | Architect | Cloud Solution Delivery Leader

Dr. Gail Zhou is a hands-on technology leader with extensive experience in software engineering, enterprise architecture, and cloud solution delivery. Known for turning complex customer and platform opportunities into practical cloud solutions that help customers adopt Azure, Fabric, AI, and data platforms with confidence.

Deep expertise in infrastructure as code, data and AI platforms, and application development. Trusted for systems thinking and pragmatic leadership, with 9.5+ years of Microsoft Azure experience, including AI Foundry and Fabric platforms. Led 60+ customer engagements that exceeded expectations. Industry-spanning experience across High Tech, Financial Services, Telecom, Government, Healthcare, Retail, and Pharma.

Awarded two U.S. patents for innovation in wireless applications, smart messaging and logging. Published author (12 Repositories) on GitHub and Microsoft Azure Architecture Center (1 Article).

Experience Summary:

- 9.5 years at Microsoft, leading customer engagements and open-source solution accelerator delivery:
 - ✓ Led delivery and publication of 13 open-source projects, including high-utilization GitHub repositories used by Microsoft field teams, partners, and customers.
 - ✓ Led 60+ customer engagements across industries, guiding cloud migration, expansion, and optimization with a focus on simplicity, automation, and repeatable success. Representative customers include Bank of America, Sony Pictures, Anthem, PepsiCo, Walgreens, Eli Lilly, Neogrid, Coca-Cola, Heineken, EY, Chubb, MasterCard, the U.S. Army, and the Government of Canada.
- 5 years of consulting experience with TCS and NTT Data, serving representative customers including the U.S. Air Force, World Bank Group, New York Life, and Citibank.
- 5 years in enterprise architecture leadership roles at Truist, Bank of America, and Fiserv.
- 7 years of software engineering and architecture design at AT&T.

Additional information is available through my public GitHub pages:

- Intro: <https://github.com/docgailzhou/introduction>.
- Resume: <https://github.com/introduction/bio/resume/DrGailZhouResume.pdf>.
- Bio: <https://github.com/docgailzhou/introduction/bio/DocGailZhouIntro.pdf>.
- GitHub repository list and speaking-event list: <https://github.com/docgailzhou/introduction/publications>.

Solution lead and key contributor to GitHub repositories and engineering publications:

1. **Fabric IQ lead and key contributor (2026):** Microsoft IQ Gold Standard Solution Accelerator (GSA) ([Microsoft-IQ-GSA](#)) or <https://aka.ms/MSIQ>.
2. **Solution Lead & Key Contributor (2026):** Fabric Real Time Intelligence Operations GSA ([Fabric-RTI-Ops](#))
3. **Solution Lead & Key Contributor (2025):** Unified Data Foundation with Fabric GSA ([UDF](#)).
4. **WAF-Aligned and Sandbox Solution Deployments (2025):** BICEP (GitHub: [ModernizeCodeSA](#)).
5. **Solution Lead:** RFP Agent (2025) (GitHub: [Agent-for-RFP-Response-SA](#)).
6. **Solution Lead (2024):** Task Management in Sustainability Manager GSA (GitHub: [TaskMgmtSA](#)).
7. **Solution Lead (2024):** Comparative Analysis for Sustainability GSA (GitHub: [CompAnalysisSA](#)).
8. **Solution Lead and key contributor (2023):** Intelligent Field Service Solution Accelerator (GitHub: [IntelligentFieldServiceSA](#)).
9. **Solution Lead and Key Contributor (2022):** Automate PDF Forms Processing ([AutomatePDF](#)).

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10. **Solution Lead and Key Contributor (2022):** Azure PDF Form Processing Automation Solution Accelerator (GitHub: [AzurePDFAutomationSA](#), YouTube: [Azure PDF Proc Automation](#)).
11. **Solution Lead and Key Contributor (2022):** Azure Invoice Process Automation Solution Accelerator (GitHub: [AzureInvoiceProcAutomation](#)).
12. **Solution Lead and Key Contributor (2020):** Azure Synapse Scripts and Accelerators ([SynapseAccelerators](#)). Process and Scripts to automate SQL Server to Azure Synapse Migration ([SQL2Synapse Link](#)) with training videos ([YouTube Link](#)).
13. **Solution Lead and Key Contributor (2018):** APS to Azure Synapse Migration ([APS2SynapseVersion1](#)), with a Python program that translates APS code to run in Azure Synapse.

CONFERENCES / EVENT SPEAKING / PUBLICATIONS

- 2026: Microsoft Internal Learning Session: AI-Aided Architecture Design
- 2026: Microsoft Internal Learning Session: AI-Aided Reusable and Customizable Sample Data Generation.
- 2025: LinkedIn Article: [A reusable .NET-based Azure solution framework](#).
- 2024: LinkedIn Article: [Information Processing and Coding in the Age of AI](#).
- 2024: Microsoft Internal Learning Session: Microsoft Fabric.
- 2023: Microsoft Customer and Partner Solutions Conference: IoT Telemetry Anomaly Detection in D365 Field Services.
- 2022: Microsoft AI/ML Airlift Conference: Azure PDF Processing Automation using AI/ML.
- 2021: PASS Summit: SQL Server to Azure Synapse Migration.
- 2021: Microsoft GBB Airlift: Business Case Development and Presales Process.
- 2021: Microsoft Modern Data Warehousing in the Cloud Digital Event.
- 2019: Microsoft Azure Air-Lift Conference Speaker on Azure Synapse Analytics.

EMPLOYMENT HISTORY

Microsoft, Principal Software Engineer / Principal Architect / Architect

01/2017–Present

Career Highlights: Better Together Award in May 2019, Special Award in December 2021, promotion to Principal Architect in 2022, and Principal Software Engineer role since 2024.

- Principal Software Engineer, Commercial Solutions Areas CTO Engineering, March 2024 – Present
 - ✓ Solution Lead: Microsoft IQ Gold Standard Solution Accelerator (GSA) ([Microsoft-IQ-GSA](#))
 - ✓ Solution Lead: Fabric Real Time Intelligence Operations GSA ([Fabric-RTI-Ops](#))
 - ✓ Solution Lead: Unified Data Foundation with Fabric GSA ([UDF](#)).
 - ✓ Infrastructure and security lead engineer: Bicep delivery for sandbox and WAF-aligned solution architectures (GitHub Repo: [Modernize Your Code Solution Accelerator](#)).
 - ✓ Solution Lead: Led engineering delivery for sustainability and Copilot patterns, promoting engineering best practices, repeatable processes, quality solutions, and clear documentation. GitHub repositories include [Agent for RFP Response SA](#), [Comparative Analysis for Sustainability SA](#), and [Task Management for Sustainability Manager SA](#).
- Principal Architect, Industry Solution Delivery and Early Access Engineering, Dec 2021 – Mar 2024. Promoted to **Principal Architect** in March 2022.
 - ✓ Led the development and publication of intelligent field service solutions using Microsoft Dynamics 365 connected services and IoT technologies. The solution is used to remotely monitor and manage IoT devices. (GitHub: [IntelligentFieldServiceSA](#)).

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- ✓ Authored and published Microsoft engineering article: [AutomatePDF](#).
- ✓ Led the development and publication of Azure PDF Form Processing Automation Solution Accelerator (GitHub: [AzurePDFAutomationSA](#); YouTube: [Azure PDF Proc Automation](#)).
- ✓ Led the development and publication of “Azure Invoice Process Automation Solution Accelerator” (GitHub: [AzureInvoiceProcAutomation](#)).
- ✓ Continued to support previously published solution accelerators and services, and created various statements of work (SOWs).
- Lead Architect, Analytics and AI Accelerate Program, Dec 2020 – Mar 2022. Received **Special Award**.
 - ✓ Advised customers on target architecture and SMP/MPP/Cloud DW migrations into Azure Synapse Analytics from Teradata, Netezza, Exadata, SQL Server, APS, RedShift, and Google Big Query.
 - ✓ Led the development and publication of SQL Server to Azure Synapse Migration Accelerators ([SQL2Synapse](#)) and YouTube [Data Migration Channel](#).
- Architect, Azure Synapse Customer Success Engineering Program, Jan 2017 – Dec 2020. Received **Better Together Award**.
 - ✓ Led customer engagements with architecture design, prototyping, and production implementations for migrating on-premises SMP/MPP data platforms to Azure Synapse Analytics. 2 customers became public references: Walgreens ([Case Study Link](#)) and Neogrid.
 - ✓ Guided customers to modernize analytical data architecture with Azure Data Lake, ADF, Azure Databricks, Azure Synapse, Azure Cosmos DB, and Power BI.
 - ✓ Led the development and publication of the APS to Azure Synapse Migration Accelerators ([APS2SynapseVersion1](#)), which became the go-to toolset for Microsoft, partners, and customers.

Tata Consultancy Services (TCS), Managing Consultant, Technology Strategy and Enterprise Architecture
2014 – 2016

Career Highlights: Achieved 99–100% customer satisfaction across all projects, many leading to follow-on engagements.

- **Cathay United Bank:** Defined the enterprise architecture vision, methodology, and 5-year roadmap to support CUB’s business transformation program.
- **World Bank Group:** Delivered five strategic programs; established integrated data architecture, technology strategy, and standards for the insurance division CIO.
- **Citi Bank:** Led a mobile resiliency assessment that laid the foundation for institutionalizing resiliency requirements, patterns, and governance in the SDLC.
- **Gilead Sciences:** Delivered target information reference architecture, BI patterns, and best practices in collaboration with the BI CoE and data governance council.

Fiserv, Lead Enterprise Information Architect (Director)
2010 – 2013

Career Highlights: Received Fiserv’s highest recognition, the Center Stage Award, for delivery excellence. Created a reusable middle-tier platform, Enterprise Transaction Information Management Systems (ETIMS), that enabled new customer products, internal capabilities, and streamlined payment processing.

- Developed transformational enterprise reference architectures and defined technology standards for information management. Founded and run the *Data Governance Committee*.
- Designed logical, physical, and hot-hot high availability architectures for large-scale, real-time payment processing—used in production for fraud detection and prevention.
- Led requirements and architecture for secure mobile payments, merchant offers, and prepaid card solutions.

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OTHER EMPLOYMENT HISTORY ABBREVIATED

1995 – 2009

- **Bank of America, SVP, Lead Business Architect & Strategist** 2007-2009
Career Highlights: Three spirit medallion awards, one patent filed, and gold level volunteer. Named by YWCA as one of the Women of Achievement from Bank of America (May 2008).
- **Truist, VP, Enterprise Architect** 2006-2007
Career Highlights: Led Multi-Channel Modernization Program Architecture, including Master Data Management, Online, Retail Banking Center.
- **COX Enterprise (Manheim), Principal Consultant, BI Architect** 2004 - 2006
Career Highlights: Started a new product for inventory optimization (POINT) which was later acquired by General Motors from Manheim.
- **NTT Data Americas, Chief Architect, Program Manager** 2001 - 2004
Career Highlights: Promotion to corporate architect and member of architecture leadership council. Led large programs (\$5-140M) and delivered quality systems within budget and timeline.
- **AT&T Architect to Director of Enterprise Architecture & Other Roles** 1995 - 2001
Career Highlights: Seven progressive promotions. I started as Sr. Software Engineer, other positions include as Software Architect, Program Architect, Project Lead, eCommerce Acting Director, Technology Strategist, Director of Enterprise Application and Data Architecture.

MISCELLANEOUS

- U.S. Citizen, Bilingual (English and Chinese)
- Three U.S. patents in the areas of wireless applications, smart messaging and logging, and systems resiliency. 2 awarded (7,139,722 & 6,925,586)

EDUCATION

- MBA, Executive Management and Finance, Kennesaw State University (Employer AT&T Sponsored Program 1997-1999)
- PhD, Electrical and Computer Engineering, University of Tennessee, PhD Dissertation (AI/ML Application in Process Control): [Automatic design of autotuners for PID controllers](#).